0. **Code:** vemuriu6191 vemuriu6191.zip

1. min max/Program.cs
2. selection statements/Program.cs
3. sentinel control loop/Program.cs

1. **min max/Program.cs**

```
using System.Reflection.Metadata.Ecma335;

namespace min_max
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int currentNum = 0, prevNum = 0, maxNum = int.MinValue,
              minNum = int.MaxValue;
            bool isfirst = true, isMaxSet = false, isMinSet = false;
            Console.WriteLine("Enter sequence of integers");
            while (currentNum != 0)
            {
                Console.WriteLine("Enter a number:");
                currentNum = Convert.ToInt32(Console.ReadLine());
                if (isfirst)
                {
                    isfirst = false;
                    prevNum = currentNum;
                    continue;
                }
                if (currentNum == prevNum)
                {
                    Console.WriteLine($"Max with increasing last digit:
                      {maxNum}");
                    Console.WriteLine($"Min with decreasing last digit;
                      {minNum}");
                    Console.WriteLine($"Number of values between max and min:
                      {Countvalues(maxNum, minNum)}");
                    break;
```

int position = 0;

max position = min position = 0;

true : we are not exitting if we enter a zero.

C1

Using isMaxSet and isMinSet.

if(isMaxSet && isMinSet){

if(min position ≠ max position){

1. **min max/Program.cs**

```
            }
            if (currentNum % 10 > prevNum % 10 && currentNum > maxNum)
            {
                maxNum = currentNum;
                isMaxSet = true;
            }
            if (currentNum % 10 < prevNum % 10 && currentNum < minNum)
            {
                minNum = currentNum;
                isMinSet = true;
            }
            prevNum = currentNum;

        }

    }
    static int Countvalues(int maxNum, int minNum)
    {
        int count = 0;
        int currentNum = minNum + 1;
        while (currentNum < maxNum)
        {
            count++;
            currentNum++;
        }
        return count;
    }

}
}
```

C2

maxposition = position

C3

minposition = position

C1

isMinSet & isMaxSet are not used!

Should be maxposition, minposition

Can be calculated numerically, analytically without a loop.

This function's objective doesn't match your invocation's arguments!

2. selection statements/Program.cs

```
namespace selection_statements
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("Enter num1 :");
            int num1 = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter num2 :");
            int num2 = Convert.ToInt32(Console.ReadLine());
            Console.WriteLine("Enter num3 :");
            int num3 = Convert.ToInt32(Console.ReadLine());

            int sum = 0;
            int count = 0;
            if (num2 ≠0 && num1/num2 ≠0){
            if (num1 % num2 ≠ 0)
            {
                sum += num1 % num2;
                count++;
            }  }        if(num3≠0 && num1/num3 ≠0){
            if (num1 % num3 == 0)
            {
                sum += num1 % num3;
                count++;
            }     }        if(num1 ≠0 && num2/num1 ≠0){
            if (num2 % num1 ≠ 0)
            {
                sum += num2 % num1;
                count++;
            }          }
```

missing

A1

2. **selection statements/Program.cs** if(num3≠0 && num2/num3≠0)

```
            if (num2 % num3 ≠ 0)
            {
                sum += num2 % num3;
                count++;
            }
            if (num3 % num1 ≠ 0)
            {
                sum += num3 % num1;
                count++;
            }
            if (num3 % num2 ≠ 0)
            {
                sum += num3 % num2;
                count++;
            }
            if (count > 0)
            {
                double average = (double) sum / count;
                Console.WriteLine($"The average is {average}");
            }
            else
            {
                Console.WriteLine("The average cannot be calculated.");
            }
        }
    }
}
```

A3

} if (num1 ≠ 0 && num3/num1 ≠ 0)

A3

} if(num2 ≠ 0 && num3/num2 ≠ 0)

A3

}

A4

A2 conditions are missing.

3. **sentinel control loop/Program.cs**

INCONSISTENCIES.

You checked for the remainders in problem A, but not in problem B.

You checked for quotients in problem B but not in problem A...

```
using System.Runtime.Serialization;

namespace sentinel_control_loop
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int sum = 0;
            int count = 0;
            Console.WriteLine("Enter a pair of integers:");
            while (true)
            {
                int a = int.Parse(Console.ReadLine());


                int b = int.Parse(Console.ReadLine());
                if (a == 0 && b == 0)
                    break;
                if (b ≠ 0)
                {
                    int quotient = a / b;
                    if (quotient ≠ 0)
                    {
                        int remainder = a % b;
                        sum += remainder;
                        count++;
                        Console.WriteLine($"{a}%{b}={remainder}");
                    }
                }
                Console.WriteLine("Enter a pair of inetegrs");
```

B1

not need due to break

B2

missing condition

if(remainder ≠ 0){

}

B3

where's the other pair where quotient = b/a ???

repeat for quotient = b/a!

3. sentinel control loop/Program.cs

```
            }
            if (count > 0)
            {
                double average = (double) sum / count;
                Console.WriteLine($"The average is {average}");

            }
            else
            {
                Console.WriteLine("The average cannot be calculated.");
            }
        }
    }
}
```

B4